The main goal of this section is to prove the following fact:

Theorem 18.1

If G is a finite abelian group then G is isomorphic to a direct product of cyclic groups whose orders are powers of primes:

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \ldots \times \mathbb{Z}_{p_k^{r_k}}$$

for primes $p_1, \ldots, p_k$ (not necessarily distinct) and integers $r_1, \ldots, r_k \geq 1$ such that $p_1^{r_1} \cdot p_2^{r_2} \cdot \ldots \cdot p_k^{r_k} = |G|$.

Example. Take $72 = 2^3 \cdot 3^2$. Theorem 18.1 says that every abelian group of order 72 is isomorphic to one of the following groups:

$$\begin{array}{ll} \mathbb{Z}_8 \times \mathbb{Z}_9 & \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 & \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \end{array}$$

The proof of theorem Theorem 18.1 will be split into two steps:

1) First we will show that if G is an finite abelian group then $G \cong G_1 \times \ldots \times G_k$ where $|G_i| = p_i^{r_i}$ for some prime p_i and $r_i \geq 1$ (Corollary 18.4).

2) Then we will prove that if the order of an abelian group G is a power of some prime, then G is isomorphic to a direct product of cyclic groups (Corollary 18.6).

The tool used in both parts if the following fact.

Theorem 18.2

Let G be an abelian group, and let H, K be subgroups of G such that

1) $H \cap K = \{e\}$
2) for every $g \in G$ there exist $h \in H$, $k \in K$ such that $g = hk$

Then $G \cong H \times K$.

Note. Theorem 18.2 is not true in general for non-abelian groups. For example, in the dihedral group D_4 take subgroups $H = \{I, R_{90}, R_{180}, R_{270}\}$, $K = \{I, V\}$. Then both condition of Theorem 18.2 are satisfied, but $D_4 \ncong H \times K$. Indeed, $H \cong \mathbb{Z}_4$ and $K \cong \mathbb{Z}_2$, so an isomorphism $D_4 \cong H \times K$ would give an isomorphism $D_4 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$. This is impossible, since the group $\mathbb{Z}_4 \times \mathbb{Z}_2$ is abelian and D_4 is not.

Proof of Theorem 18.2. Define a function $f\colon H \times K \to G$ by $f(h,k) = hk$. We will show that this function is an isomorphism of groups.

First, we check that f is a homomorphism:

$$\begin{aligned} f((k,h)\cdot(k',h')) &= f((kk', hh)) \\ &= (kk')(hh') \\ &= (kh)(k'h') \\ &= f(k,h)\cdot f(k',h') \end{aligned}$$

By condition 2) in Theorem 18.2, the function f is onto, so it remains to show that if it is 1-1, i.e. that $\mathrm{Ker}(f) = \{(e,e)\}$. Assume then that $(h,k) \in \mathrm{Ker}(f)$. Then $hk = f(h,k) = e$ or equivalently $h = k^{-1}$. This implies that $h \in H \cap K$. By condition 2) in Theorem 18.2 we obtain that $h = e$ and so also $k = e$. □

Theorem 18.3

Let G be a finite abelian group. Assume that $|G| = p^r m$ where p is a prime, $r \geq 1$ and m is a number which is not divisible by p. Then $G \cong H \times K$ where $|H| = p^r$ and $|K| = m$.

Proof. Since p does not divide m, thus $\gcd(p^r, m) = 1$. This implies that there exist $s, t \in \mathbb{Z}$ such that $sp^r + tm = 1$.

Define subgroups $H, K \subseteq G$ as follows:

$$H := \{g \in G \mid \ g^{p^r} = e\}$$

$$K := \{g \in G \mid \ g^{m} = e\}$$

We claim these subgroups satisfy assumptions of Theorem 18.2 Indeed, if $g \in H \cap K$ then

$$g = g^1 = g^{sp^r + tm} = (g^{p^r})^s (g^m)^t = e \cdot e = e$$

Thus $H \cap K = \{e\}$. Next, take any $g \in G$. We have $g^{p^r m} = g^{|G|} = e$, This gives $(g^m)^{p^r} = e$, so $g^m \in H$ and $(g^{p^r})^m = e$, so $g^{p^r} \in K$. Denote $h = (g^m)^t \in H$ and $k = (g^{p^r})^s \in K$. We obtain

$$g = g^1 = g^{sp^r + tm} = (g^{p^r})^s (g^m)^t = kh = hk$$

By Theorem 18.2 we obtain $G \cong H \times K$.

It remains to show that $|H| = p^r$ and $|K| = m$. We have $p^r m = |G| = |H| \cdot |K|$. Since H is a p-group, by Corollary 17.8 $|H|$ is a power of p. Also, since the order of every element of K divides m and m is not divisible by p, it follows that K does not contain any elements of order p. By Cauchy Theorem 17.6 we obtain that $|K|$ is not divisible by p. Therefore $|H| = p^r$ and $|K| = m$. □

Corollary 18.4

If G is a finite abelian group and $|G| = p_1^{r_1} p_2^{r_2} \cdot \ldots \cdot p_k^{r_k}$ where $p_1, p_2, \ldots, p_k$ are distinct primes then

$$G = G_1 \times G_2 \times \ldots \times G_k$$

where $|G_i| = p_i^{r_i}$.

Proof. We use induction with respect to the number of distinct primes k. If $k = 1$ then $|G| = p_1^{r_1}$, so we take $G_1 = G$.

Assume that the statement is true for all abelian groups whose order is a product of powers of k distinct primes, and let G be a group such that

$$|G| = p_1^{r_1} p_2^{r_2} \cdot \ldots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$$

where $p_1, \ldots, p_{k+1}$ are distinct primes. By Theorem 18.3 we get $G = G_1 \times H$ where $|G_1| = p_1^{r_1}$ and $|H| = p_2^{r_2} \cdot \ldots \cdot p_k^{r_k} p_{k+1}^{r_{k+1}}$ By the inductive assumption, $H \cong G_2 \times \ldots \times G_{k+1}$ where $|G_i| = p_i^{r_i}$. This gives

$$G \cong G_1 \times H \cong G_1 \times G_2 \times \ldots \times G_{k+1}$$

□

Theorem 18.5

If G is an abelian group such that $|G| = p^n$ for some prime p and let a be the element of the largest order in G. Then there exists a subgroup $K \subseteq G$ such that

$$G \cong \langle a \rangle \times K \cong \mathbb{Z}_{p^r} \times K$$

Idea of the proof. Show that there is a subgroup $K \subseteq G$ such that $\langle a \rangle \cap K = \{e\}$ and for any $g \in G$ we have $g = a^i k$ for some $i \in \mathbb{Z}$, $k \in K$. Then use Theorem 18.2.

□

Corollary 18.6

If G is an abelian group such that $|G| = p^n$ for some prime p then G is a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p^{k_1}} \times \mathbb{Z}_{p^{k_2}} \times \ldots \times \mathbb{Z}_{p^{k_m}}$$

for some $k_1, k_2, \ldots, k_m$.

Proof. It follows from Theorem 18.5 by induction with respect to n. □

Proof of Theorem 18.1. It follows from Corollary 18.4 and Corollary 18.6. □